

Class 09: Candy Mini Project

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Background

Today we are taking a small detour to analyze a dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA)

Data Import

The data is all related to halloween candy is from the 538 website.

```
candy<-read.csv("candy-data.txt", row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0

Almond Joy	1	0	0	1	0	0
	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

There are 85 different types of candy in this dataset.

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

There are 38 fruity candy types in this dataset.

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

```
candy |>
  filter(row.names(candy)=="Air Heads") |>
  select(winpercent, pricepercent, sugarpercent)
```

```
      winpercent pricepercent sugarpercent
Air Heads  52.34146         0.511         0.906
```

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

There is a useful `skim()` function in the **skimr** package that can help give you a quick overview.

```
library("skimr")
skimr::skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

There are two columns that work differently than the other columns: The Histogram and Complete_Rate column look to be on a different scale.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

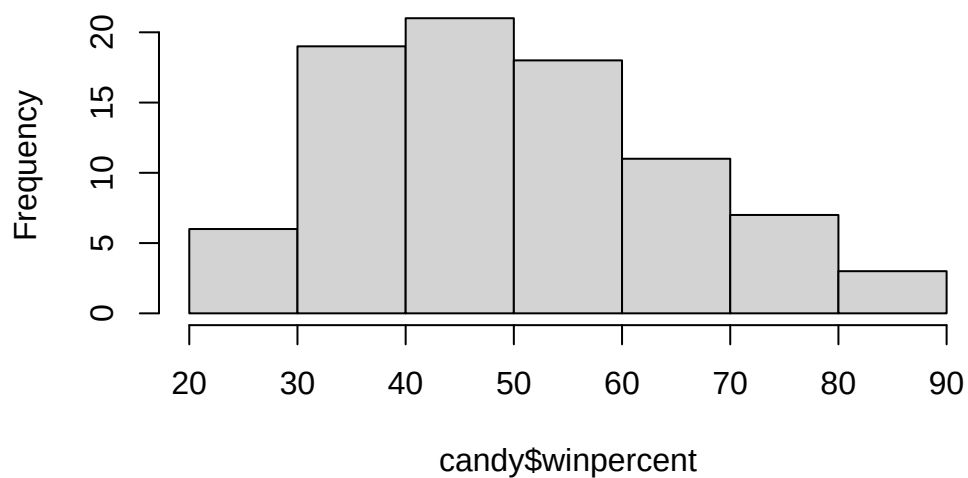
I think a one or zero represents if the candy contains chocolate or not.

Exploratory Analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
hist(candy$winpercent)
```

Histogram of candy\$winpercent



```
library(ggplot2)
ggplot(candy)+aes(winpercent)+geom_histogram(bins = 20,col="blue",fill="green")
```



Q9. Is the distribution of winpercent values symmetrical?

No, the distribution is not symmetrical. It appears to be right skewed.

Q10. Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
mean(candy$winpercent[candy$chocolate == 1])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[candy$fruity == 1])
```

```
[1] 44.11974
```

Q12. Is this difference statistically significant?

```
t.test(winpercent ~ chocolate, data = candy)
```

Welch Two Sample t-test

data: winpercent by chocolate

t = -7.3031, df = 67.539, p-value = 4.164e-10

alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:

-23.91110 -13.64744

sample estimates:

mean in group 0 mean in group 1

42.14226 60.92153

Yes, the difference is statistically significant because the p-value is much less than 0.05.

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
candy |> arrange(winpercent) |> select(winpercent) |> head(5)
```

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

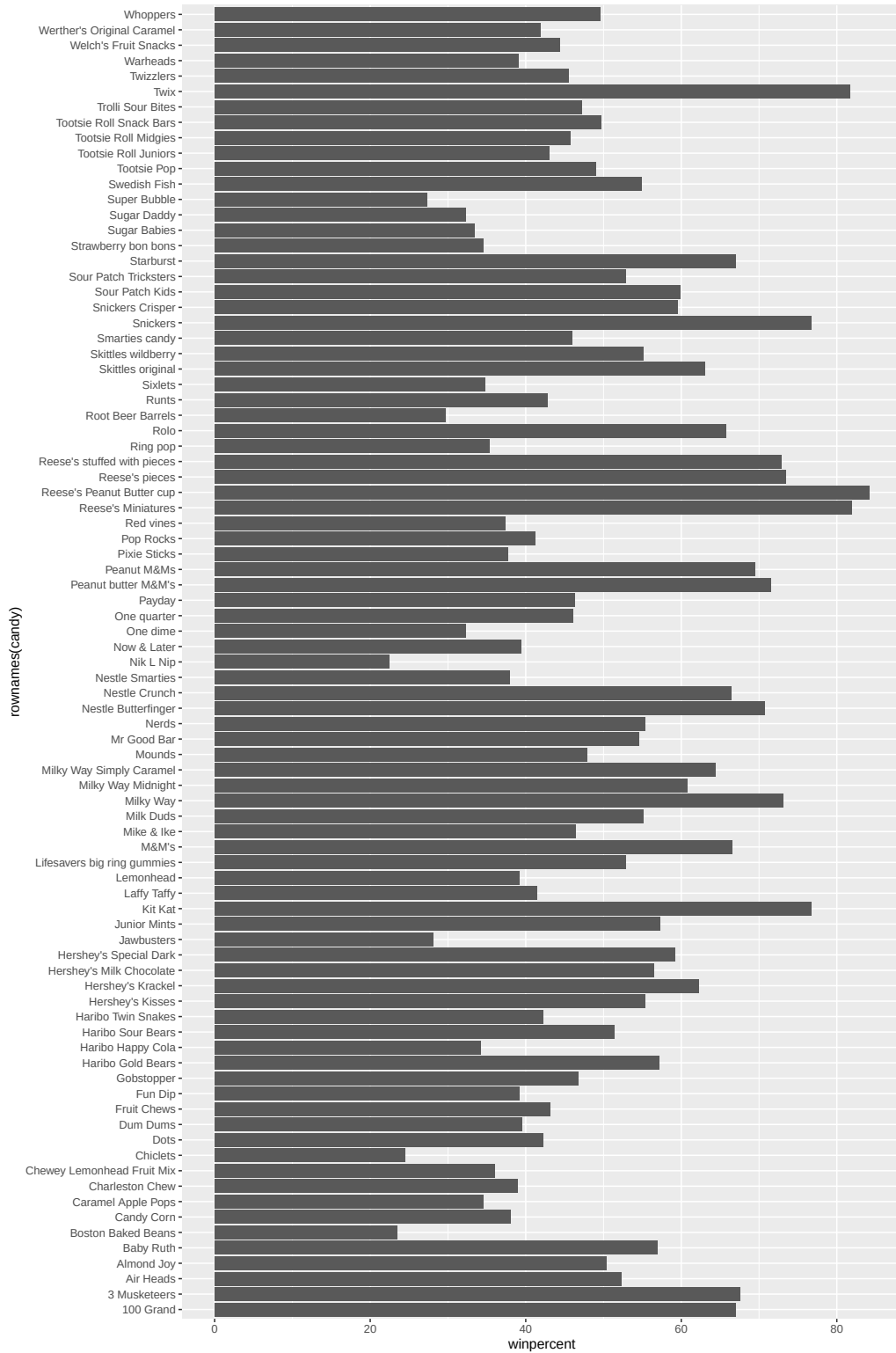
Q14. What are the top 5 all time favorite candy types out of this set?

```
candy |> arrange(desc(winpercent)) |> select(winpercent) |> head(5)
```

	winpercent
Reese's Peanut Butter cup	84.18029
Reese's Miniatures	81.86626
Twix	81.64291
Kit Kat	76.76860
Snickers	76.67378

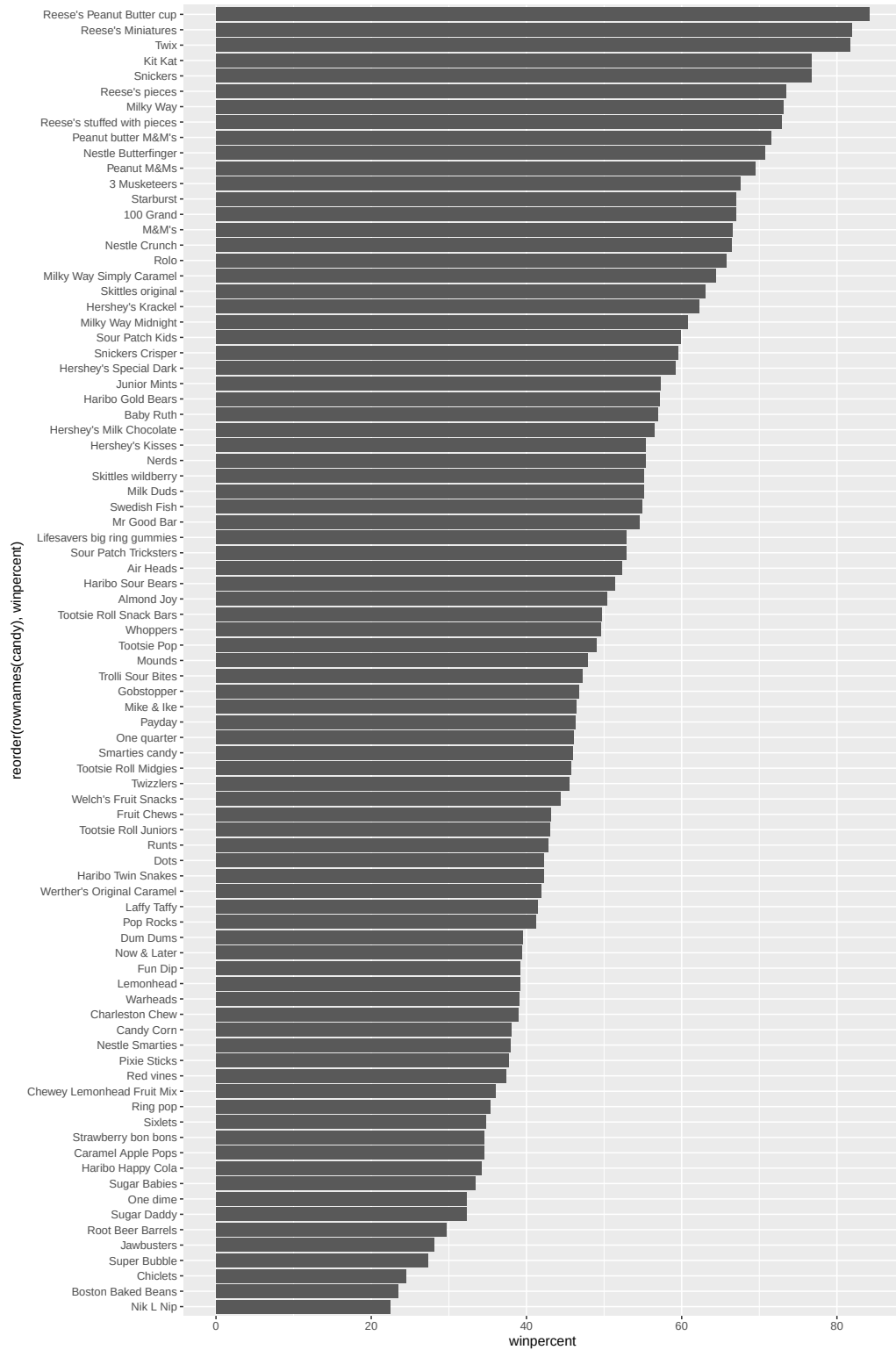
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```



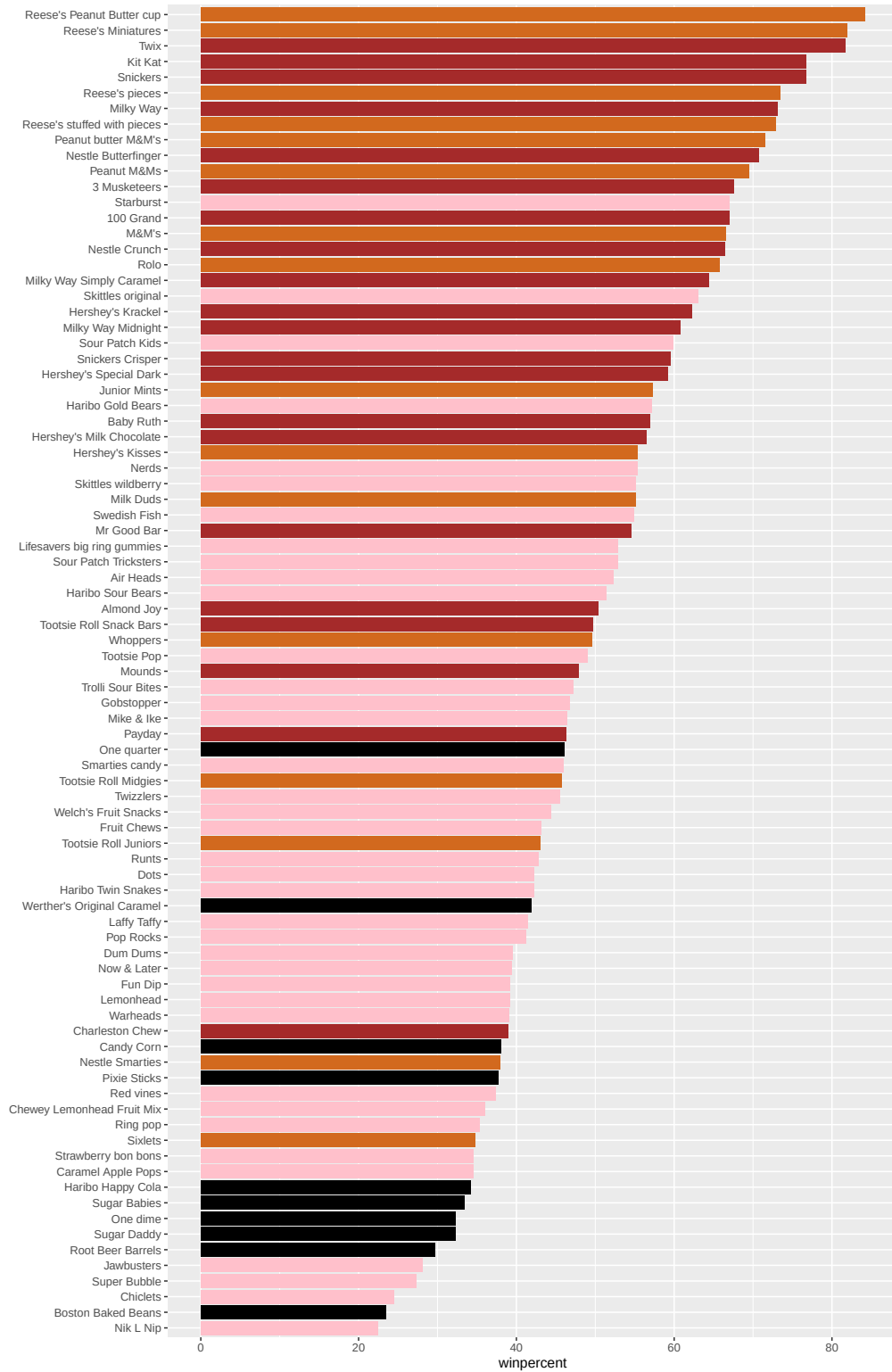
Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +  
  geom_col()
```



```
my_cols = rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```



Q17. What is the worst ranked chocolate candy?

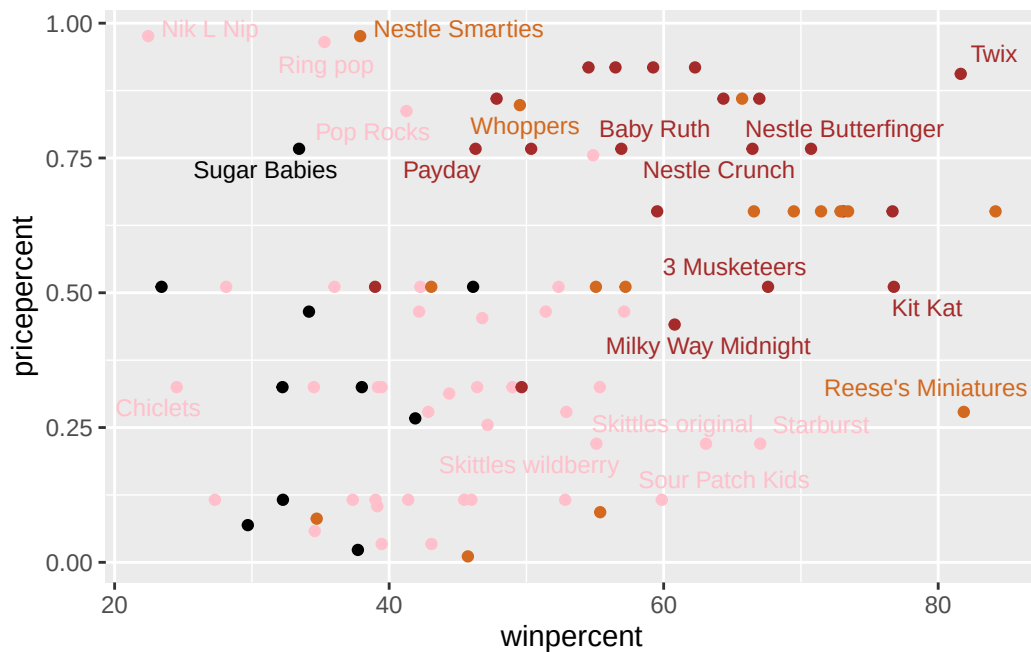
The worst ranked chocolate candy is Sixlets.

Q18. What is the best ranked fruity candy?

The best ranked fruity candy is Starburst.

Taking a look at pricepercent

```
library(ggrepel)
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's miniatures appear to offer the most bang for your buck.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

Nik L Nip, Nestle Smarties, Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate appear to be the most expensive candies in this data set. The least popular one being Nik L Nip.

```
candy |> arrange(desc(pricepercent)) |> select(pricepercent) |> head(5)
```

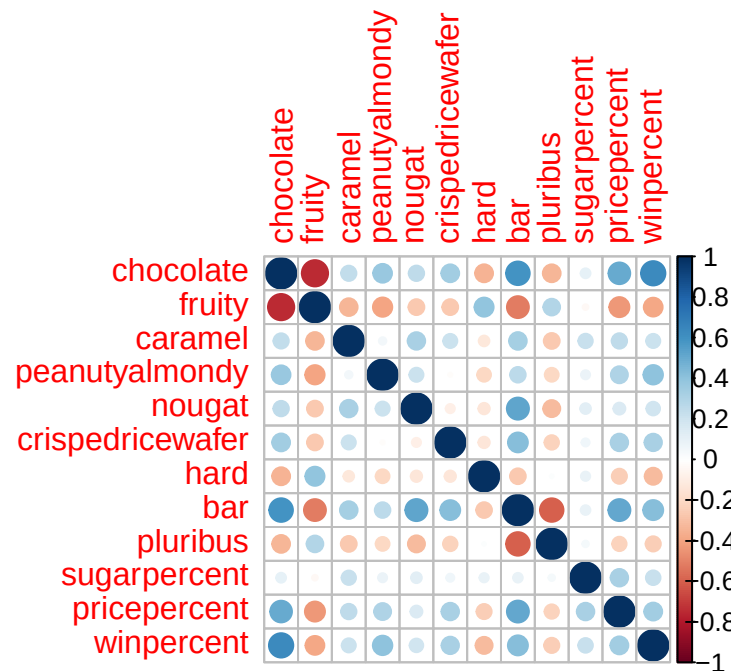
	pricepercent
Nik L Nip	0.976
Nestle Smarties	0.976
Ring pop	0.965
Hershey's Krackel	0.918
Hershey's Milk Chocolate	0.918

Exploring the correlation structure

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and Fruity are the two variables that are the most anti-correlated.

Q23. Use your corplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

From the plot, I think chocolate and fruity will have the largest contribution because they're the most anti-correlated variables on the plot and have the most negative values.

Principal Component Analysis

In this case we want to be sure to set `scale=TRUE()` argument for `prcomp()` because we have one variable, `winpercent()`, that is on a very different scale than all others and would otherwise dominate our PCA results.

```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

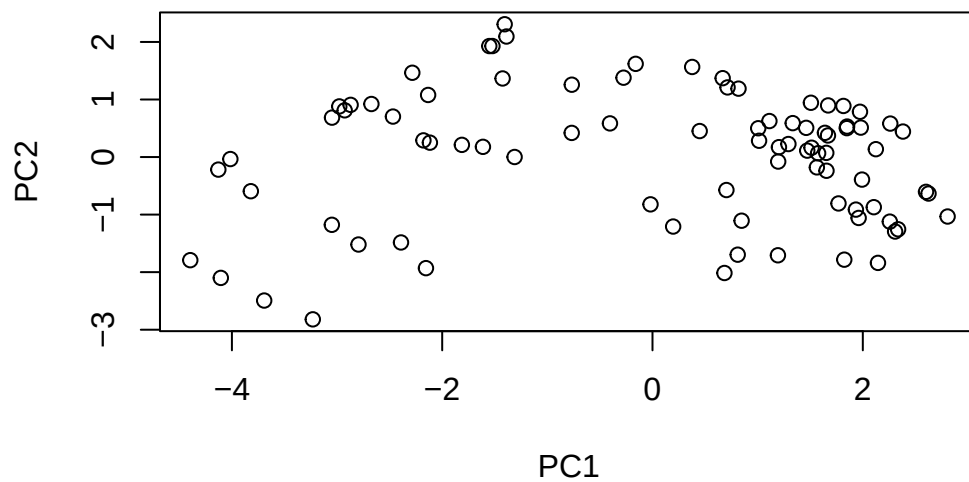
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

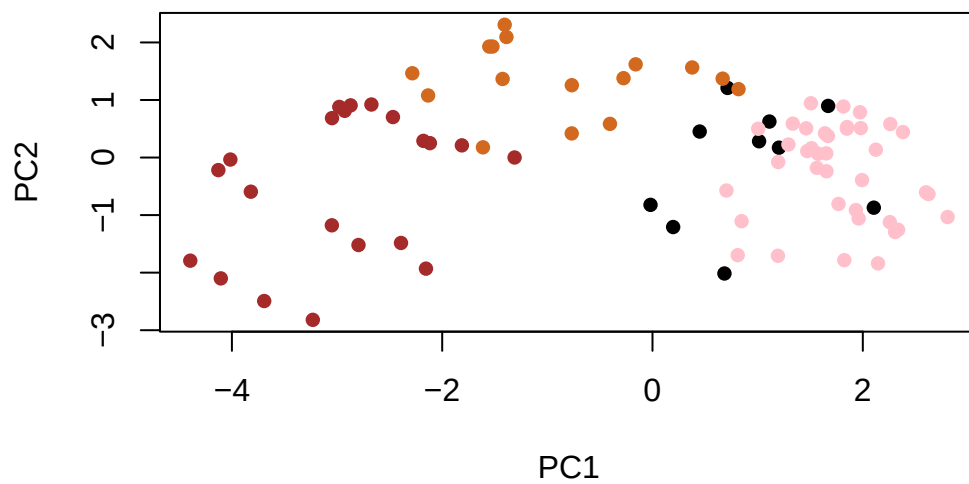
	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

First major result figure in the “score plot” of PC1 vs PC2 - how different candy are related to each other on our new PC axis:

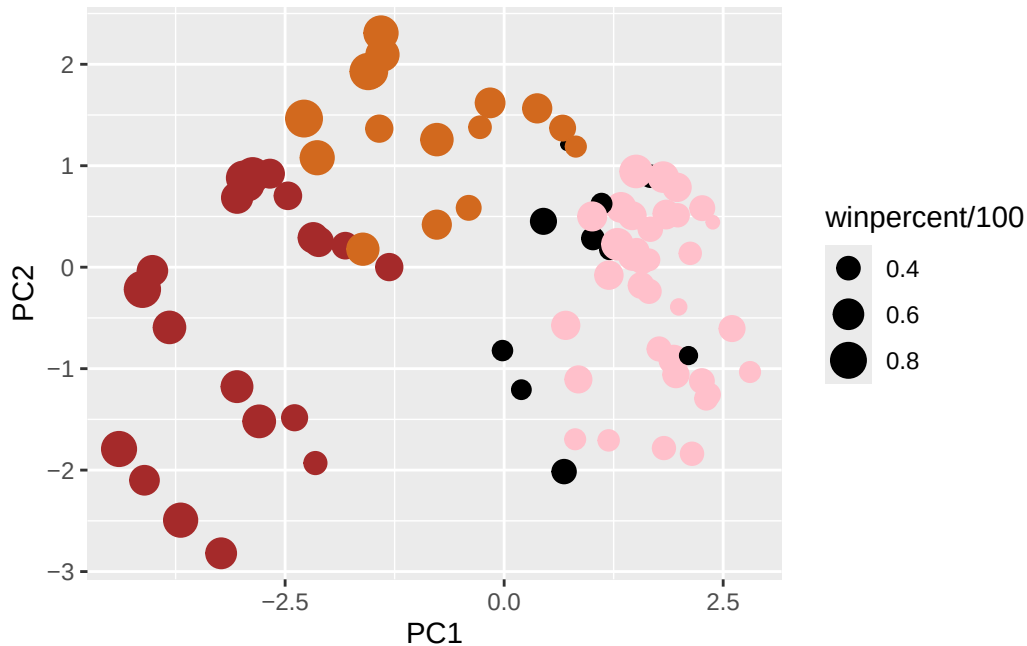
```
plot(pca$x[, 1:2])
```



```
plot(pca$x[,1:2], col=my_cols, pch=16)
```




```
my_data <- cbind(candy, pca$x[,1:3])
p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=my_cols)
p
```

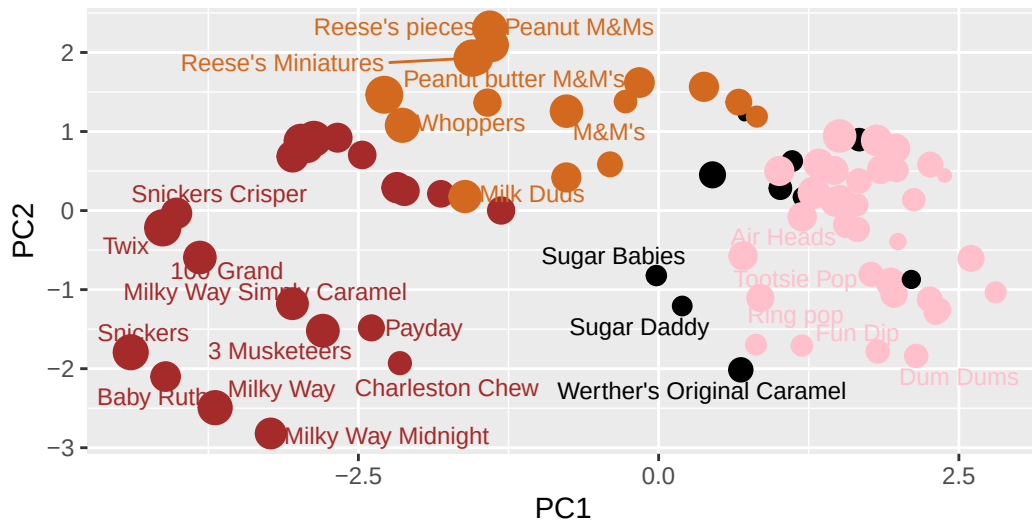


```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
       caption="Data from 538")
```

Halloween Candy PCA Space

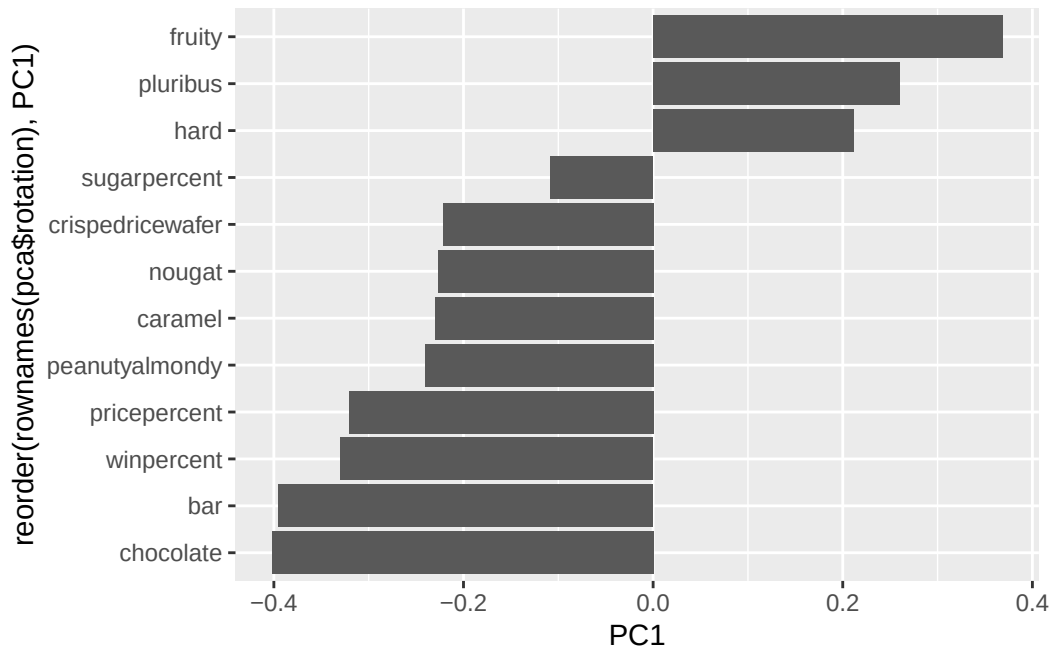
Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
library(plotly) # ruined my PDF so I hid the result.  
ggplotly(p)
```

```
ggplot(pca$rotation) +  
  aes(PC1, reorder(rownames(pca$rotation), PC1)) +  
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The variables `fruity`, `pluribus`, and `hard` are picked up strongly by PC1 in the positive direction. We saw these relationships in the `corrplot` previously where these variables have positive correlations.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Based on the results of this lab, a “winning” candy’s characteristics appear to be a chocolate bar with a higher price point, as there was correlation between the price and the popularity. Additional characteristics like peanut-almondy or crisped rice wafer also appear to contribute to a winning candy. The barplot visualization allowed me to see what types of candy were the most popular, which was Reese’s chocolate. The correlation helped me identify which characteristics are positively correlated with `winpercent`. Then lastly, the PCA showed that chocolate bar candy with additional features were in the same direction as `winpercent`, which confirmed which characteristics make up a winning candy.